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Twitter Sentiment Analysis as a Predictor of Voting Patterns in the Eurovision Song Contest

Project Proposal

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1. Abstract

This project aims to explore the power of Twitter opinion mining in predicting real life voting patterns in the Eurovision Song Contest.

Can a sentiment analysis of tweets relating to the Eurovision Song Contest be used to predict the voting behaviour of groups? If so, can sentiment strength predict the strength of the overall results? Is Twitter sentiment more in line with the public vote or with the jury decisions? While some research has been conducted into these questions this project will go further, asking if other factors, such as cultural and geographical proximity or past voting behaviour can be combined with sentiment analysis to better predict the outcome.

This project will deliver a data flow for the aggregation and predictive analysis of tweets related to the competition. The collected data will be used to explore the ways in which opinion mining of Twitter data can be used to model and predict real-life events.

2. Introduction

2.1 Problem Statement

How can opinion mining of Twitter help to predict the results of the Eurovision Song Contest?

2.2 Objectives

The microblogging platform Twitter is a ubiquitous part of modern day life. Used by corporations, politicians, public figures and the general public to promote themselves, share information and engage with each other. The huge amount of data generated, an average of 6,000 tweets per second [internetlivestats.com, 2018], is a gift to data scientists.

This project will aim to design and implement a data flow for the aggregation and predictive analysis of tweets related to the Eurovision Song Contest. I plan to use this data to explore the ways in which a sentiment analysis of tweets can reflect real life group behaviours. As an event, Eurovision is worthy of study as it has an inherent element of crowdsourcing; those tweeting can affect the result by telephone voting and a tweet can have an influence on the voting behaviour of others, potentially affecting the outcome. Analysing the sentiment of tweets has been used to predict the outcomes of sporting events and elections, however this project's focus will be one of the most watched and most tweeted about television events of the year, decided in part by public vote: the Eurovision Song Contest.

This project will attempt to answer the following detailed questions:

i. Can a sentiment analysis of tweets relating to the Eurovision Song Contest be used to predict the outcome? If so, can sentiment strength predict strength of results? Is Twitter sentiment more in line with the televote (see section 3.1) or with jury decisions? Is Twitter sentiment a good predictor of real-life crowd behaviour?

ii. Can other factors be combined with sentiment analysis to better predict the outcome? Can cultural and geographical proximity or past voting behaviour, for example, improve the prediction?

iii. What are the best methods for answering the above questions?

As I discovered in my review of the available literature, past research projects on Twitter sentiment analysis have usually focused on sports or politics and not on cultural or entertainment events.

3. Background Research

3.1 Eurovision and the Scoring System

The Eurovision Song Contest (commonly referred to as Eurovision) is a musical competition held each year between members of the European Broadcasting Union (EBU). First held in 1956 among seven countries, the 2018 competition is due to be contested by 43 countries [Eurovision.tv, 2018]. Since 2004, Semi-finals have been held in the week leading up to the Grand Final to reduce the number of competitors to a manageable number for a single broadcast; the host country automatically qualifies for the final along with the 'Big Five' countries who make the greatest financial contribution to the EBU.

The contest is decided by a combination of public televotes and jury votes. Since 2016, the results from each country's jury are announced by a representative. The top ten countries are awarded points; 12 points for first place, 10 points for second, and 8 points down to 1 for third to tenth places (figure 1). Points for the televote, where the public call or text a telephone number to register their support for a country's performance, are calculated in the same way but are added together and announced in one block. This means each voting country gives out 116 points, with a maximum of 24 to one single country should the jury and televote choose the same song in first place. The Grand Final and Semi-finals are all decided in this way but only those taking part in a Semi-final are able to vote. The host country and 'Big Five', not included in either of the Semi-finals, are allocated one in which they can vote. No country may award points to itself.

Figure 1 - Presenter before announcing the UK's 1st and 2nd choices at the 2015 competition. Numbers in white circles indicate points awarded by the UK. Numbers in white text indicate cumulative total.

3.2 Literature Review

Social media as a predictor of real life events is a well-studied phenomenon. Shoen et al. [2013] describe three methods of prediction using social media: prediction market models, or the wisdom of crowds, which use the opinions of large numbers of people to predict future events; survey models, in which a random sample is surveyed and behavioural intentions are used to predict future events; and statistical models, where past events are analysed statistically to predict future events. For this project prediction market models would be the most relevant method of prediction using sentiment analysis while statistical models could be used to see if past voting behaviour has any effect on future voting behaviour. Gayo-Avello [2011], however, cautions against embracing Twitter too enthusiastically as a predictive tool. He studied predictions made by researchers about the results of the 2008 United States presidential election from Twitter data, including some that employed opinion mining techniques, and argues that they minimised the importance of demographic bias, employed naïve sentiment analysis methods and were too eager to declare that past events would predict future outcomes.

Data sets of tweets lend themselves to sentiment analysis – they are necessarily short, typically written by a human and they often contain unfiltered opinion. Thelwall et al. [2011] developed an algorithm, SentiStrength, which allocates both a positive and negative sentiment score to a piece of text based on, among other things, the words included, punctuation and the presence and nature of intentional misspellings. The dual scores reflect the possibility of a sentence containing both positive and negative sentiments; “I miss you” for instance, suggests both love and longing, representing positive and negative feeling simultaneously. The accuracy achieved by SentiStrength when analysing English language short social texts is human-level [Thelwall et al., 2010].

Some research has been conducted into how opinion mining can be used to interpret the results of Eurovision. Kakouris et al. [2016] found some correlation between Twitter sentiment and competition performance. Here the Twitter data set was created at multiple points in advance of the competition and analysed using a Bag of Words method of opinion mining and frequent term analysis. Although not studying the content of tweets, Highfield et al. [2013] visualised the network of Eurovision fans and found some interesting results; the use of the official hashtags #Eurovision and #ESC split Europe into two factions (figure 2). They also highlight the importance of the official accounts @eurovision and @bbceurovision, around which activity centred. Pavlyshenko [2013] collected tweets the day before the 2013 Grand Final and created association rules using keywords such as “love” and “winner”. They found a strong link between opinion mined from Twitter and performance in the competition which they argue is unsurprising as those sending tweets can have a direct impact on the outcome of the competition by participating in the televote, unlike in similar events such as

sporting fixtures. Brevik and Selmer [2013] achieved an accuracy of only 50% in predicting the top ten countries using a sentiment analysis of tweets. One issue they highlighted is that of language; Ireland, as an English speaking country, placed first in their prediction when in reality they came last, possibly because their methods heavily favoured countries who were tweeted about in English.

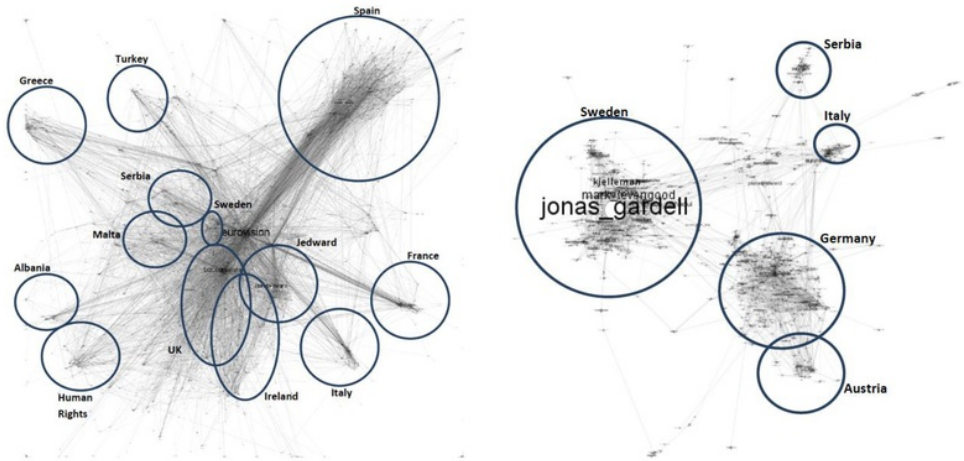

Figure 2 – comparison of #Eurovision and #ESC network maps, 22-28 May 2012, degree range of 10 or more [Highfield et al. 2012].

Further research has been conducted into why countries vote the way they do and what this can tell us about modern Europe but without a focus on Twitter behaviour. Fenn et al [2005] studied competition voting behaviour from 1998 to 2004 and used a cluster analysis to find groups of countries who have historically displayed similar voting behaviour, finding that such groups do exist; while some follow geographic, cultural and historical proximity (e.g. UK and Ireland) others are not so clear cut (e.g. Portugal were more closely aligned with France than Spain). Baio and Blangiardo [2013] modelled past votes using Bayesian hierarchical modelling and found voting patterns "based on cultural and geographical similarities, as well as migrations of people" but that these patterns were relatively weak.

4. Project Details

4.1 Approach and Methodology

The project will be executed in a fashion similar to the main Twitter analytics projects seen in the literature review; an agile programming methodology will lead to the organisation of available data sources, analytics and visualisation. Data preparation, filtering and aggregation will be designed and implemented from scratch using the Python programming language as this is a suitable language for data analytics. To carry out this project a high level of proficiency in Python is required as well as knowledge of statistical tools and data analysis techniques.

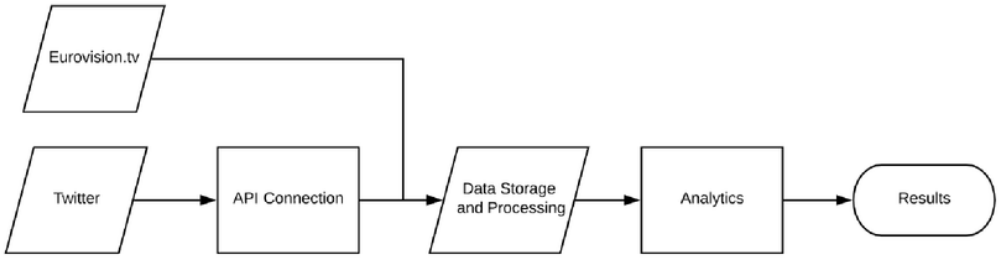

Figure 3 – Proposed data flow.

4.2 Data Gathering

There are two sets of data that must be used to analyse this problem. The first, the results data, are released each year at Eurovision.tv by the EBU after the contest. This shows points awarded by each voting country to each competing country as well as the breakdown between each voting country's jury votes and televotes and scores submitted by individual jurors (figure 4).

Televoting

Countries	Total	Cyprus	Greece	Portugal	Romania	Sweden	FYR Macedonia
Israel	**5**	0	0	0	0	0	1
Poland	**41**	0	0	0	0	5	0
Belarus	**33**	0	0	0	0	0	0
Austria	**0**	0	0	0	0	0	0
Armenia	**21**	1	2	0	0	0	0
Netherlands	**15**	0	0	0	0	0	0
Moldova	**264**	6	6	12	12	8	2

Figure 4 – small selection of the data made available on Eurovision.tv after 2017's competition. Combined, juror and telephone votes are provided separately (https://eurovision.tv/history/full-split-results).

The second set is a selection of tweets relating to the competition. Using Twitter's Streaming Application Programming Interface (API) a program will be written that downloads and stores tweets containing chosen keywords and their metadata live (figure 5). The Python package

Tweepy will be used to do this, with data initially stored in JSON format, collecting only tweets relating to the competition. The most straightforward indicator that a tweet is Eurovision-related is that it contains an official hashtag; in 2017 official hashtags included #Eurovision, #ESC2017, #CelebrateDiversity, #12Points and #douzepoints [Jordan, 2017]. Since 2015 there have also been official 'hashflags', which display a Eurovision-themed flag when a combination of letters indicating that the tweet is related to a specific country is typed as a hashtag [Storvik-Green, 2015]; these have returned for every competition since and will be useful both for data gathering and data analysis.

```
{
"place": null,
"in_reply_to_status_id_str": null,
"user": {
"followers_count": 147,
"profile_background_image_url_https": "",
"profile_link_color": "1DA1F2",
"screen_name": "p%%%%%%%",
"profile_sidebar_border_color": "C0DEED",
"geo_enabled": false,
"profile_text_color": "333333",
"id_str": "966879859665354752",
"profile_background_tile": false,
…
"full_text": "Drat, if I was a member Hungary would be 27th instead of 32nd. Guess I'm joining next year to save the graces of my faves hahaha #Eurovision @ESCFAN_Club https://t.co/QkOGT0lxGP",
…
}
```


Figure 5 – Example tweet (top) and small selection of information collected from Twitter in JSON format (bottom) (https://twitter.com/paloinje/status/984752618273587200).

The timing of the data gathering is important. Since all songs have been announced in advance and are available on Eurovision's official YouTube channel, fans, commentators and bookmakers already have their favourites meaning that Tweets collected well in advance could be useful. However, the staging and audience reaction is first seen the night before the live Semi-finals when the juries first vote. The live Semi-finals give the viewing public their first look at the performances so tweets from those nights could be useful. In the days between the Semi-finals and the Grand Final anticipation builds so this is another option. The final option is during the live Grand Final when viewers tweet along with the live show. Previous research has tended to gather tweets in advance. I would suggest collecting tweets during the live show will give the broadest demographics. If we divide watchers between Eurovision viewers, those who only watch the live Grand Final, and Eurovision fans, those who follow Eurovision news, country selections and watch the Semi-finals, it is possible casual viewers would be more likely to tweet during the final than in advance, giving a broader range of opinion.

4.3 Data Processing

There are a number of pre-processing tasks to consider when using data collected using Twitter's API; Should retweets be included? Or records with missing parameters? How should emoji be dealt with? Are there any irrelevant parameters? How should tweets that mention multiple countries be dealt with?

To extract sentiment from the collected tweets I propose to use SentiStrength, a tool created by academics at the University of Wolverhampton, explained in more detail in the literature review. This can be used to give each tweet a dual sentiment rating. A limitation of this algorithm is its focus on the English language. Tested and untested versions exist for some other languages but not every language spoken in Europe; when studying a global event such as Eurovision, reliance on one language could be problematic, as seen in Brevik and Selmer's work. However, any alternative to SentiStrength would likely suffer the same problem.

4.4 Building/Training/Testing

The two sets of data will allow me to create a model that investigates the link between Twitter sentiment and competition performance. Tweets will be grouped by origin and target countries (i.e. where the tweet originates from and who the tweet is about) and evaluated individually for sentiment using SentiStrength (figure 6). These scores will be aggregated and compared with the actual points awarded by the televote and juries. The actual scores can act as targets, enabling semi-supervised learning techniques to be employed, dividing the data into training and testing data. This is a benefit of analysing the results retrospectively.

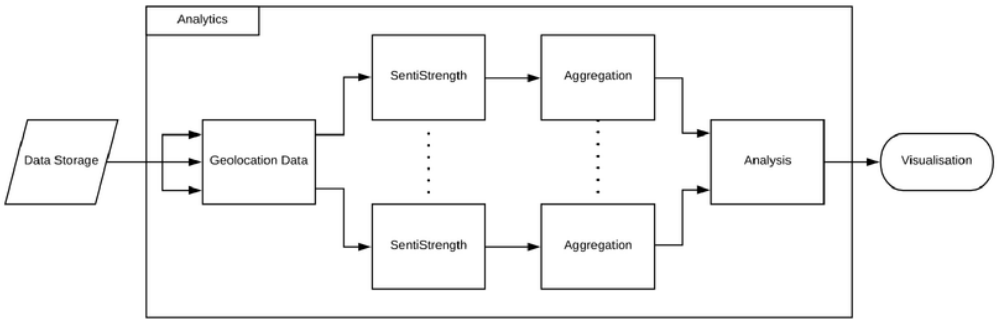

Figure 6 – structure of proposed analytical solution.

Different targets can be set (jury, televote or combined results) and the inputs can be varied to include other parameters such as geographical and cultural proximity to improve performance.

4.5 Risk Management

One risk to this approach is that the data gathering stage will happen live and could fail due to untested volumes of tweets (according to Twitter data there were 8 million on the night of the Grand Final in 2017 [Beverton-Palmer, 2017]). To mitigate this the data gathering program will be tested during the two Semi-finals. Should it fail on the night Twitter's Rest API will be used to search for tweets from the last seven days, however there are rate limits imposed that would affect the volume of data to be collected.

The suitability of the data is vital to the success of the project. The proposed method relies on recognizing the geographical origin of a tweet. Should the rate of geotagged tweets not provide sufficient data (in 2016 only 2-3% of tweets were geotagged [Dredze et al., 2016]), locations can be extracted from profile info provided by the user as well as time zones stored as metadata.

There is also the risk that a link cannot be found. However, the literature review suggests that a correlation will be found between sentiment and voting patterns. The data released by the EBU is broad enough that different avenues can be explored; among the televote results and jury vote, broken down into individual jurors, and with the addition of extra parameters, I am confident that a connection will be found.

4.5 Project Plan

This is a preliminary schedule of my work. It is important to notice that times here are heavily influenced by the need to collect data before or during the night of the Eurovision Grand Final.

Phase	Actions	Duration	Key Dates
1	Connecting to Twitter's API and data gathering	Apr 17th – May 12th	May 8th – 1st Semi-final May 10th – 2nd Semi-final May 12th – Grand Final
2	Data pre-processing and sentiment analysis	May 13th – Jun 11th	
3	Building initial models	Jun 12th – Jul 9th	
4	Training, testing, improving models	Jul 10th – Aug 20th	
5	Analysing and writing up findings	Aug 21st – Sep 10th	
6	Submitting project report	Sep 11th – Sep 17th	September 17th – submission deadline

5. Summary

The power of Twitter as a predictive tool for real life events has been the subject of some academic work, as explored in the Literature Review (Section 3.2). The microblogging platform offers a wealth of information, viewable in real time, regarding the opinions of individuals. Much of the available research focusses on sport, however Eurovision, as a much tweeted about cultural event with an element of crowdsourcing at its core lacks the same breadth of research. Previous studies analysing Twitter sentiment as a predictor of Eurovision voting patterns have had some success. This project will take that research further, looking at how information obtained via opinion mining can be combined with other factors to model voting behaviour.

The programming for each stage (data gathering, processing and building, training and testing) will be implemented using Python. The main work in the gathering stage is to scrape Twitter for tweets before or during the night of the Grand Final of the competition. The processing stage will consist of cleaning the data and assigning sentiment using SentiStrength; removing noise and irrelevant or unusable tweets. A model will be built to answer the research questions, its format and parameters adjusted to find the best method.

A successful project will deliver a data flow for the aggregation and predictive analysis of tweets related to the Eurovision Song Contest. This data will be used to model the ways in which a sentiment analysis of Twitter behaviour can reflect and predict real life events.

6. Glossary

API	application programming interface.
EBU	European Broadcasting Union.
Emoji/emoticon	a small picture representation an object or emotion (e.g. ☺ or ;-)).
Hashflag	a series of letters preceded by a hashtag that Twitter converts to a Eurovision themed flag.
Hashtag	a string preceded by '#'. Used to search for tweets related to similar topics (e.g. '#Eurovision').
Geotag	location data stored as tweet metadata.
Retweet	a repost of another user's tweet (shortened to 'RT').
The Eurovision Song Contest	a musical competition held each year primarily between members of the EBU. Popularly referred to as 'Eurovision', 'Grand Prix', or abbreviated to ESC. The competition is not limited to European countries (Israel, Australia etc.).
Tweet	a message posted on Twitter.
Twitter	a microblogging platform. Users can post 280-character messages.

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